

Task -1

Static IP Address Assignment and Connectivity Test

For this lab, I used the private network **192.168.1.0/24** and assigned five devices static IP addresses in the same subnet.

Device	Static IP Address	Subnet Mask	Default Gateway
Laptop 1	192.168.1.10	255.255.255.0	192.168.1.1
Laptop 2	192.168.1.11	255.255.255.0	192.168.1.1
Desktop 1	192.168.1.12	255.255.255.0	192.168.1.1
Desktop 2	192.168.1.13	255.255.255.0	192.168.1.1
Laptop 3	192.168.1.14	255.255.255.0	192.168.1.1

Steps

1. Connect all five devices to the same router/switch or simulated LAN.
2. Open the network settings on each device.
3. Select **IPv4 → Use the following IP address**.
4. Assign the static IP addresses shown in the table.
5. Use the same subnet mask, 255.255.255.0, on all devices.
6. Set the default gateway to 192.168.1.1.
7. Save the settings.

8. Open Command Prompt on each device and test connectivity.

For example, from **Laptop 1**:

ping 192.168.1.11

ping 192.168.1.12

ping 192.168.1.13

ping 192.168.1.14

Expected Result

A successful test should show:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Reply from 192.168.1.12: bytes=32 time<1ms TTL=128

Reply from 192.168.1.13: bytes=32 time<1ms TTL=128

Reply from 192.168.1.14: bytes=32 time<1ms TTL=128

This confirms that all five devices are in the **same subnet and can communicate with each other**.

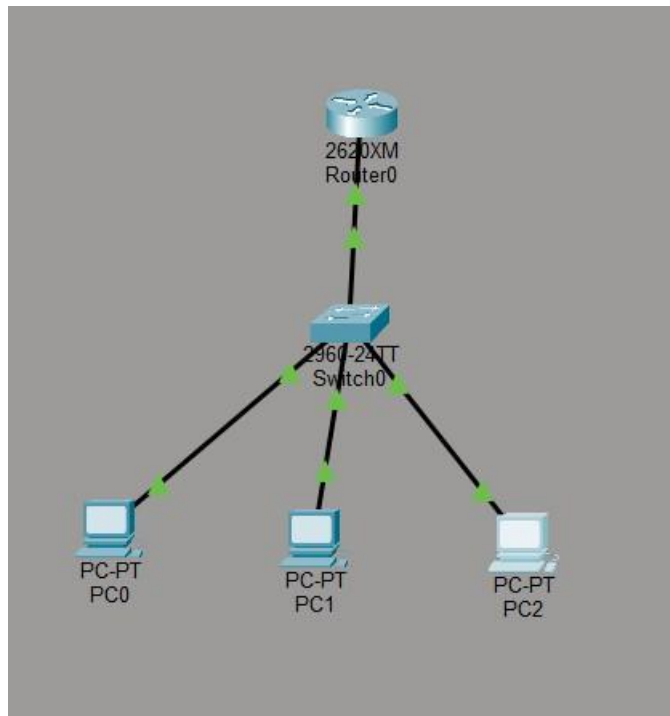
Conclusion: The five devices were assigned unique static IP addresses within the 192.168.1.0/24 subnet. Ping tests verified successful connectivity between the devices.

Task-2

Q-Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Configure a DHCP server to automatically assign IP addresses to at least three devices and verify that each device receives a unique IP address from the configured range.

Network Topology



Router DHCP Configuration:

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.20
Router(config)#ip dhcp pool LAB_POOL
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Configure the PCs For each PC:

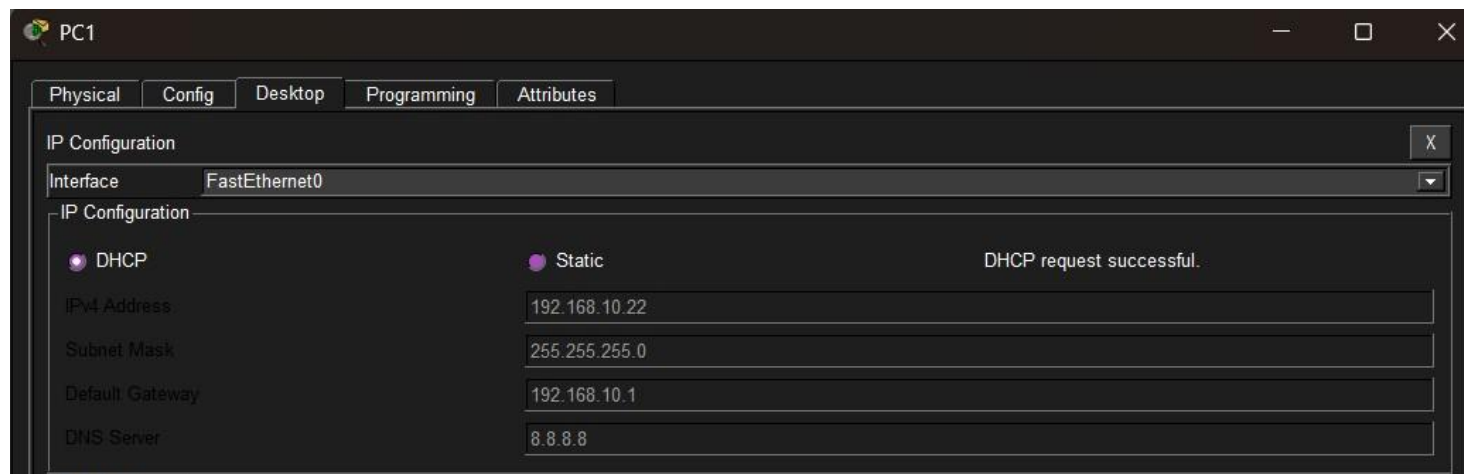
1. **Open Desktop.**
2. **Select IP Configuration.**
3. **Click DHCP.**

The PCs should automatically receive IP addresses.

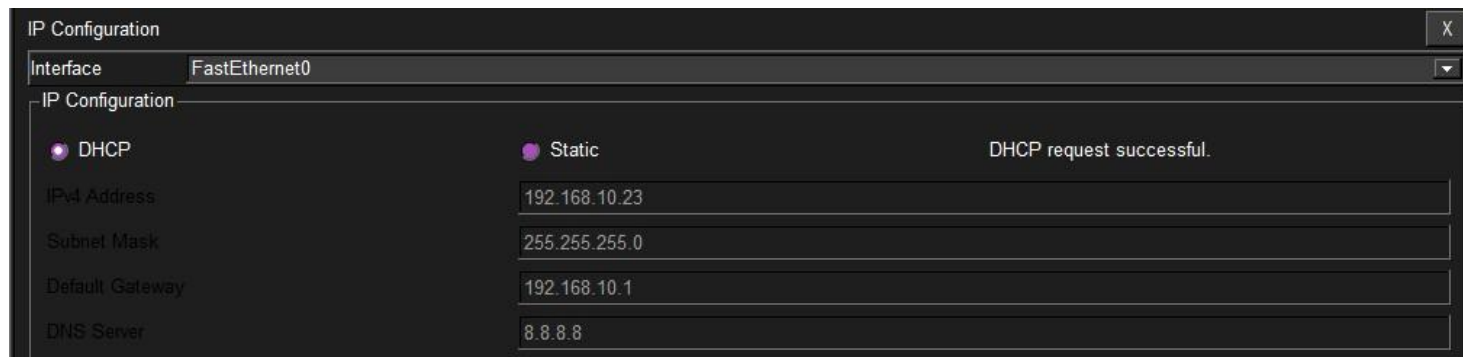
PC0:



PC 1:



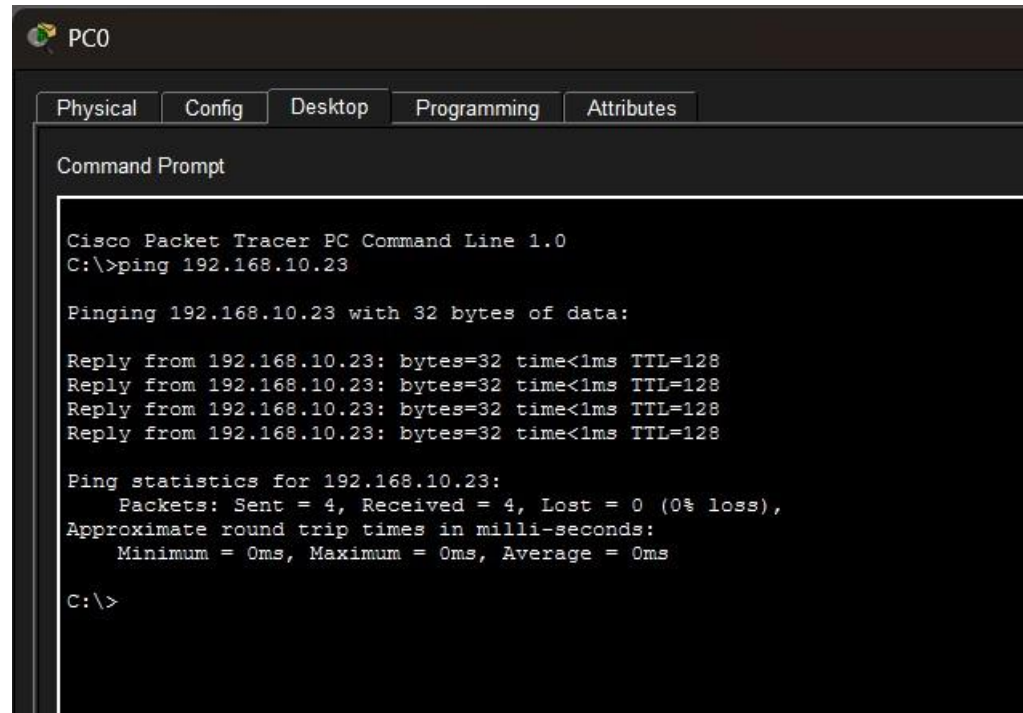
PC 2:



The image shows a network configuration window titled "IP Configuration" with a close button (X) in the top right corner. Below the title bar, there is a dropdown menu labeled "Interface" with "FastEthernet0" selected. Underneath, a section titled "IP Configuration" contains two radio buttons: "DHCP" (which is selected) and "Static". To the right of these buttons, the text "DHCP request successful." is displayed. Below the radio buttons, there are four input fields for network parameters: "IPv4 Address" with the value "192.168.10.23", "Subnet Mask" with "255.255.255.0", "Default Gateway" with "192.168.10.1", and "DNS Server" with "8.8.8.8".

Interface	Value
Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.10.23
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	8.8.8.8

Test connectivity:



The screenshot shows a PC0 window in Cisco Packet Tracer. The 'Command Prompt' tab is active. The text in the command prompt is as follows:

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.23

Pinging 192.168.10.23 with 32 bytes of data:

Reply from 192.168.10.23: bytes=32 time<1ms TTL=128
Reply from 192.168.10.23: bytes=32 time<1ms TTL=128
Reply from 192.168.10.23: bytes=32 time<1ms TTL=128
Reply from 192.168.10.23: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.23:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Task-3

To map a custom hostname (musicplayer.local) to a local IP address using the Windows hosts file and verify that the hostname opens a local web page.

Step 1: Open the Hosts File

1. Open **Notepad** as **Administrator**.
2. Click **File** → **Open**.
3. Navigate to:

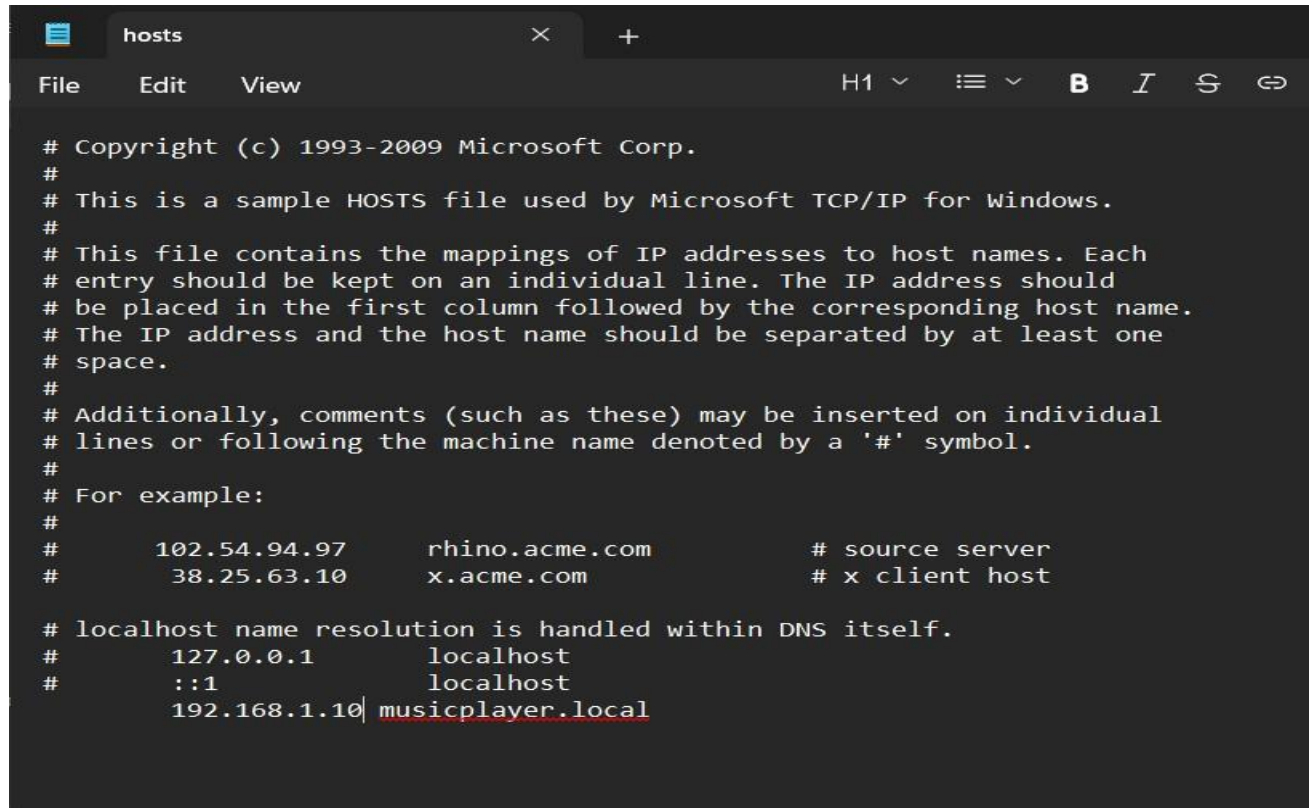
C:\Windows\System32\drivers\etc\

4. Change the file type from **Text Documents (*.txt)** to **All Files (.)**.
5. Open the **hosts** file.

Step 2: Add the Hostname Mapping

Add the following line at the end of the file:

192.168.1.10 musicplayer.local Save the file.

A screenshot of a Notepad++ window titled 'hosts'. The window has a menu bar with 'File', 'Edit', and 'View'. The right side of the menu bar contains icons for font size (H1), font style (B, I, S), and a search icon. The main text area contains the following text:

```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com       # source server
#       38.25.63.10       x.acme.com          # x client host
#
# localhost name resolution is handled within DNS itself.
#       127.0.0.1         localhost
#       ::1               localhost
#       192.168.1.10| musicplayer.local
```

Step 3: Verify the Configuration

1. Open **Command Prompt**.
2. Run the following command:

ping musicplayer.local

3. Confirm that the hostname resolves to **192.168.1.10**.

Step 4: Test in a Browser

1. Open a web browser.
2. Enter:

`http://musicplayer.local`

3. Verify that the local web page opens successfully.

Observations

Parameter	Value
Operating System	Windows
Local Web Server IP	192.168.1.10
Hostname	musicplayer.local
Hosts File Modified	Yes
Ping Test	Successful

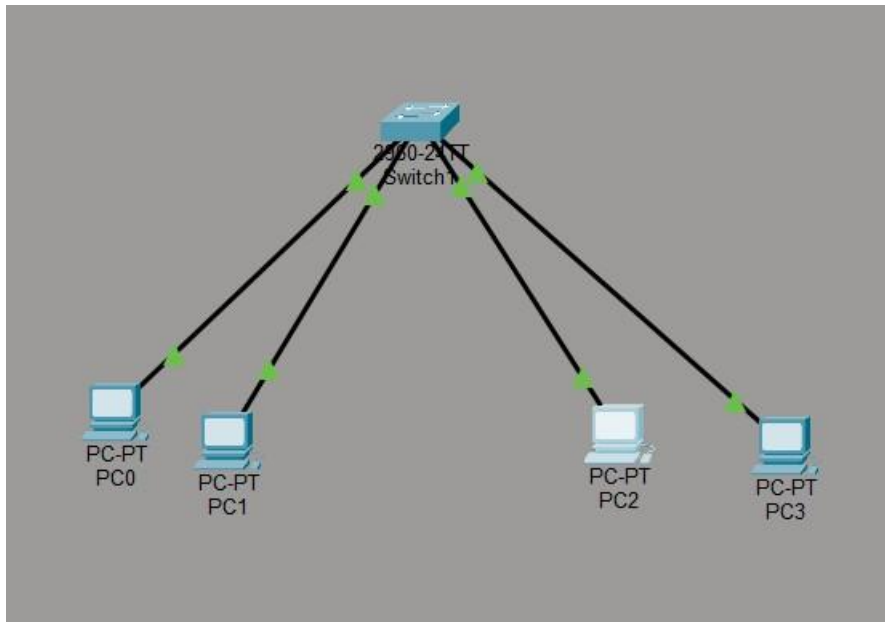
Browser Test	Web page loaded successfully
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The Windows **hosts** file was successfully configured to map the hostname **musicplayer.local** to the local IP address **192.168.1.10**. The system resolved the hostname correctly, and the local web page was successfully accessed using the custom hostname instead of the IP address.

Task-4

To create two VLANs on a switch, assign devices to each VLAN, and verify that devices within the same VLAN can communicate while communication between different VLANs is blocked.

Network Topology:



PC IP Address

Device VLAN	IP Address
PC0 VLAN 10	192.168.10.2
PC1 VLAN 10	192.168.10.3
PC2 VLAN 20	192.168.20.2
PC3 VLAN 20	192.168.20.3

Switch Configuration

```

Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Workstations
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Smart_devices
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show vlan

```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Workstations	active	Fa0/1, Fa0/2
20	Smart_devices	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0

--More--

Connectivity Test

Test 1: Same VLAN Communication From PC0:

ping 192.168.10.3 Result: **Successful** From

PC2:

ping 192.168.20.3

Result: **Successful**

Test 2: Different VLAN Communication From PC0:

ping 192.168.20.2

Result: **Request timed out** From **PC1:**

ping 192.168.20.3

Result: **Request timed out**

Two VLANs, **VLAN 10 (Workstations)** and **VLAN 20 (Smart Devices)**, were successfully created on the switch. Two devices were assigned to each VLAN. Devices within the same VLAN communicated successfully using the ping command, while communication between different VLANs was blocked because no inter-VLAN routing was configured.